Building energy study

A multi-family building in Danderyd

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1 What the brief asks for

A housing association has one multi-family building that cannot keep 20 °C indoors when the weather is really cold. The brief asks for the radiator power on a design day, the balance temperature, the average U-value, the heating season in the city's weather file, and the energy class once that year has been brought back to a normal climate.

The birth date is 29 August 1986. Written as year, month, day (1986, 08, 29) it gives $a = 0$, $b = 8$, $c = 2$, $d = 9$, $ab = 8$, $cd = 29$, the same split as the example in the brief (1985, 10, 29 gives $a = 1$, $b = 0$, $c = 2$, $d = 9$). Indoor setpoint $T_{\text{in}} = 20 + c = 22.0$ °C. The heating system is an air-to-air heat pump, COP 3.

Table 1: Answers the brief asks for.

Quantity	Value
City	Danderyd (number 29)
DVUT	−10.7 °C (7-day)
Indoor temperature T_{in}	22.0 °C
Average heat transfer coefficient U_m	0.589 W/(m^2·K)
Balance temperature T_b	18.21 °C
Heating power, no internal gains	46.5 kW
Same power, appliance gains only	41.1 kW
Heat gain, people	3.8 kW, 22.2 MWh
Heat gain, appliances	5.4 kW, 31.5 MWh
Solar heat gain	19.9 MWh
Heat loss, transmission	109.0 MWh
Heat loss, ventilation	43.6 MWh
Heat deficit, weather year 2017	80.2 MWh
Degree hours, base T_{in}	107 242 °C·h
Heat deficit, normal-year correction	81.7 MWh
Primary energy number	108.3 kWh/m^2·year
Energy class	E

Seasonal energies in Table 1 cover the heating season (15 September to 15 May). The duration diagram of the outdoor temperature is Figure 2.

2 The building

Three storeys $(3+a)$, plan 10×45 m. Heated floor area $A_{\text{temp}} = 1\,350$ m^2. Storey height 2.5 m, so the heated volume is 3 375 m^3. The brief gives the plan as 10×45 m^2 and no internal dimensions, so A_{temp} is that plan times the number of storeys.

Gross façade 825 m^2, of which windows 129.0 m^2 and doors 50.4 m^2. Opaque wall left: 645.6 m^2. The enclosing area A_{om} is 1 725 m^2 (roof + ground floor + whole façade, windows and doors included).

External corridors run along both long sides, on the intermediate floors. With two intermediate floors and two sides of 45 m, the thermal-bridge length is 180 m at $\Psi = 0.6$ W/(m·K).

3 U-value of the outer wall

The brief asks for the λ-value method. The studs are 45 mm wide at 600 mm centres, so the wood fraction is $45/600 = 0.075$. Mineral wool is 0.037 W/(m·K) and timber is 0.14 W/(m·K):

$$\lambda_{\text{mix}} = 0.925 \times 0.037 + 0.075 \times 0.14 = 0.0447 \text{ W/(m·K)}. \tag{1}$$

The ventilated cavity and the outer leaf (the brick) are given as one resistance, 0.20 m^2·K/W. Asbestos and the plastic foil are ignored, as the brief says. Indoor surface resistance is the usual 0.13 m^2·K/W for a wall. Gypsum is 13 mm at 0.25 W/(m·K).

$$R = 0.13 + \frac{0.013}{0.25} + \frac{0.095}{\lambda_{\text{mix}}} + 0.20 = 2.506 \text{ m}^2\text{·K/W}, \tag{2}$$

so $U_{\text{wall}} = 1/R = 0.399$ W/(m^2·K). Horizontal noggins are not given, so they are not included. Counting them would raise U a little.

4 Average heat transfer coefficient

U_m is the transmission heat loss, bridges included, divided by A_{om}. Ventilation is not part of U_m.

Table 2: Transmission coefficients. Windows are double glazing in timber frames.

Part	UA or ΨL (W/K)
Opaque walls	257.6
Windows, $U = 2.9$ W/(m^2·K)	374.1
Doors, $U = 2.0$ W/(m^2·K)	100.8
Roof, $U = 0.16$ W/(m^2·K)	72.0
Ground floor, $U = 0.23$ W/(m^2·K)	103.5
Balcony slabs, ΨL	108.0
Total H_{tr}	1 016.0

$$U_m = H_{\text{tr}}/A_{\text{om}} = 1\,016.0/1\,725 = 0.589 \text{ W/(m}^2\text{·K)}. \tag{3}$$

A new multi-family building in BBR is limited to 0.40 W/(m^2·K). This envelope is above that limit. The windows alone are a large part of H_{tr}.

5 Ventilation

Thirty-eight people is $30 + ab$. Each produces 0.005 L/s of CO_2, outdoor air is 350 ppm, and the indoor limit used for sizing is 1 000 ppm. Parts per million is a volume fraction, so the difference is multiplied by 10^{-6}.

$$q = \frac{38 \times 0.005 \text{ L/s}}{(1\,000 - 350) \times 10^{-6}} = 292 \text{ L/s} = 0.22 \text{ L/(s·m}^2\text{)}. \tag{4}$$

Leakage is 0.05 air changes per hour, 47 L/s. The system is mechanical exhaust, so there is no heat recovery. For a design-day calculation the two flows are added: that is the cold-day case, outdoor air coming in both through the vents and through the cracks.

$$H_{\text{vent}} = 1.2 \text{ kg/m}^3 \times 1\,000 \text{ J/(kg·K)} \times q = 407.0 \text{ W/K}. \tag{5}$$

The specific flow from the CO_2 balance is below the 0.35 L/(s·m^2) often used as a hygienic floor for housing. The brief says to size from the CO_2 measurement, so the lower flow is kept. Raising it to 0.35 would increase the ventilation loss, and the heating power with it.

6 Time constant and DVUT

DVUT is not the coldest hour of one year. It is the outdoor temperature of a cold spell whose length matches the building's time constant. Only the mass on the warm side of the insulation counts, and only the first 10 cm from each indoor face (ISO 13790). The brick sits outside the insulation, so it is left out. Using the whole 200 mm ground slab would pretend that the soil-side concrete follows the room temperature.

The active capacity is $C = 822$ MJ/K. With $H = H_{\text{tr}} + H_{\text{vent}}$:

$$\tau = C/H = 6.69 \text{ days}. \tag{6}$$

Rounded to the nearest tabulated duration, that is the 7-day DVUT for Danderyd: −10.7 °C (Boverket / SMHI, 1991–2020).

7 Balance temperature and design power

T_b is the outdoor temperature at which appliance gains alone cover the losses. People and sun are left out, as the brief asks for T_b and for the design power.

$$T_b = T_{\text{in}} - \frac{Q_{\text{appliances}}}{H_{\text{tr}} + H_{\text{vent}}} = 22.0 - \frac{5\,400}{1\,423} = 18.21 \text{ °C}. \tag{7}$$

Above T_b the building does not need the radiators, if the appliances are running and sun and people are still ignored. Below T_b it does.

The question on power without internal heat generation is the full loss at DVUT. The hint then says to keep the appliances. Both numbers are reported. The safe size of the heating system is the larger one: appliances are not a guarantee at six in the morning.

$$\begin{aligned} P &= (H_{\text{tr}} + H_{\text{vent}}) \times (T_{\text{in}} - \text{DVUT}) = 46.5 \text{ kW}, \\ P - Q_{\text{appliances}} &= 41.1 \text{ kW}. \end{aligned} \tag{8}$$

On that design day the loss splits as in Figure 1 and Table 3.

Table 3: Design-day heat loss at DVUT, $T_{\text{in}} = 22$ °C.

Path	Power (kW)
Ventilation and leaks	13.3
Windows	12.2
Outer walls	8.4
Balcony slabs	3.5
Ground floor	3.4
Doors	3.3
Roof	2.4

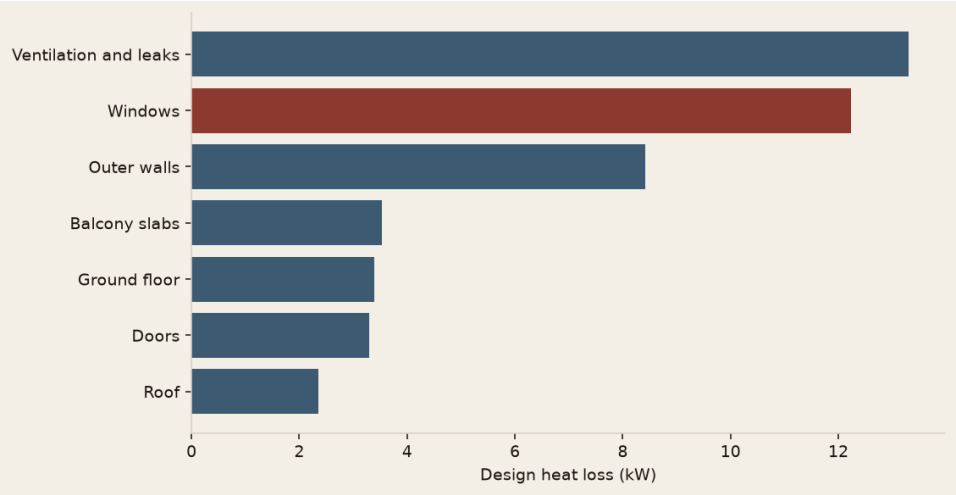


Figure 1: Where the design-day heat leaves the building. Ventilation is slightly ahead of the windows.

8 Heating season, 15 September to 15 May

The weather file is Sveby / SMHI for Danderyd, 2017 ($2015 + a + a + c$). Solar radiation is the monthly curve printed in the brief (Örebro, 60° N), used for every city as instructed. It was read from the printed figure to about ±5 W/m^2. The values are monthly averages, so they are applied to every hour of the month and then multiplied by the hours, as the hint in the brief says.

People contribute 3.8 kW: 38 people × 100 W. The CO_2 rate in the brief, 0.005 L/s per person, is about one met, which is about 100 W of heat. Presence is taken as continuous. A real schedule, with people out during the day, would lower this gain.

Each hour,

$$Q_{\text{heat}} = \max\bigl(0,\; H_{\text{tr}}(T_{\text{in}} - T_{\text{out}}) + H_{\text{vent}}(T_{\text{in}} - T_{\text{out}}) - Q_{\text{app}} - Q_{\text{people}} - Q_{\text{solar}}\bigr). \tag{9}$$

Surplus in a mild hour is not saved for the next hour. There is no utilisation factor: gains are fully useful until they match the loss. That is slightly optimistic.

The season is 5 832 hours. Degree hours with base T_{in}: 107 242 °C·h. Degree hours with base T_b, used for the climate correction: 85 117 °C·h in 2017 and 86 725 °C·h in the typical year.

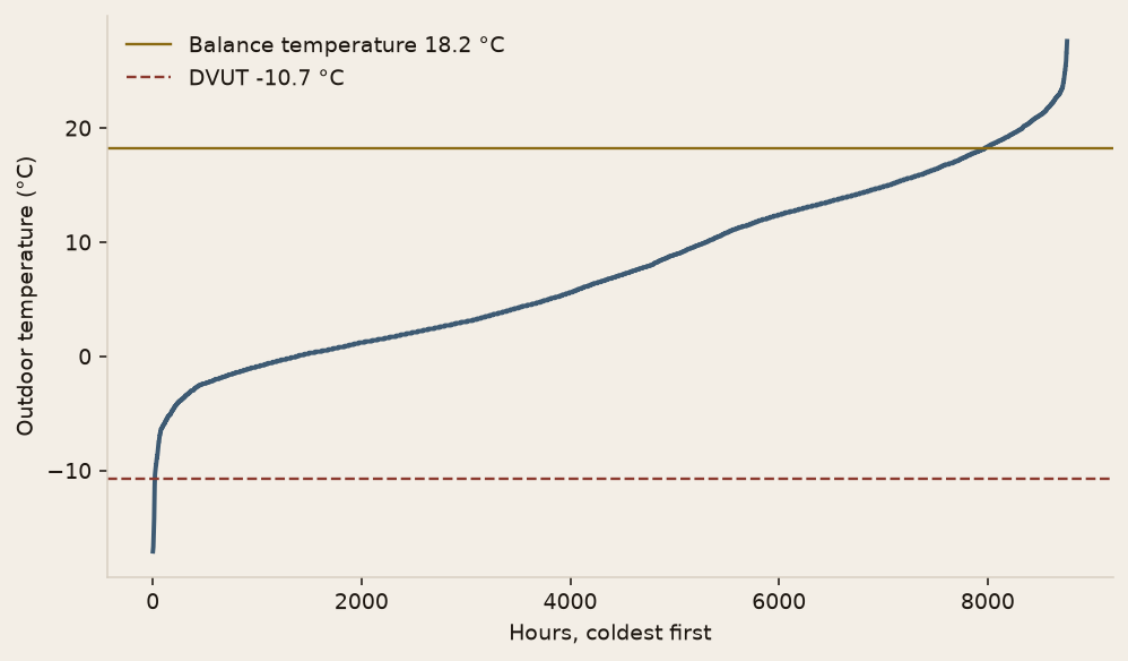


Figure 2: Outdoor temperature in Danderyd, all hours of 2017, sorted from cold to mild. Solid line: balance temperature. Dashed line: DVUT.

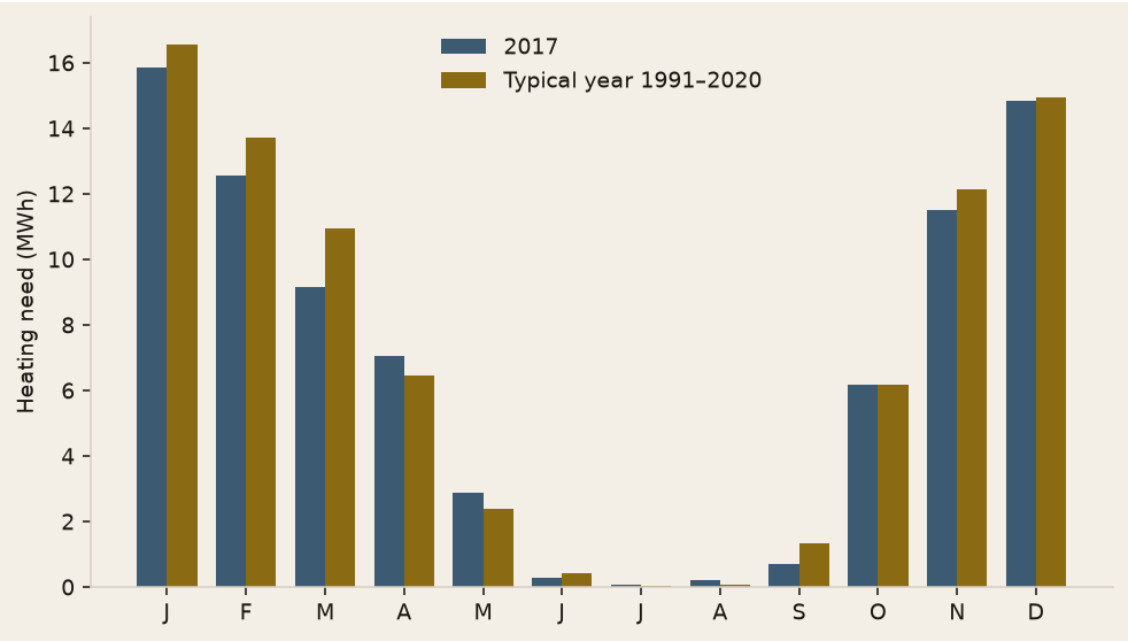


Figure 3: Heating need by calendar month. 2017 beside the typical year 1991–2020. Summer months stay near zero.

9 Normal year and energy class

The lecture correction scales the climate-dependent heat with the ratio of degree hours (base T_b) between the typical year and 2017:

$$E_{\text{normal}} = E_{2017} \times \frac{\text{degree hours}_{\text{normal}}}{\text{degree hours}_{2017}} = 80.21 \text{ MWh} \times 1.019 = 81.72 \text{ MWh}. \tag{10}$$

Running the same hourly balance on the Sveby typical year (1991–2020) gives 83.08 MWh directly. The two routes are close. The energy class below uses the scaled 2017 result, which is the correction the brief points to. The direct run is the check.

The heating system is an air-to-air heat pump with COP = 3. Electricity bought for space heat is the heat divided by 3. An air-to-air unit does not heat domestic hot water, so hot water is counted as direct electric (25 kWh/m^2, Sveby). Property electricity (fans, common lighting) is 15 kWh/m^2, also Sveby, also electricity. Household appliances are already inside the heat balance as a gain; their electricity is not part of the building's primary energy number.

Weighting factors are the BBR ones: electricity 1.8, district heating would have been 0.7. Only space heating is divided by the geographical factor F_{geo}. For Danderyd, $F_{\text{geo}} = 1.0$, so the factor does not move the number.

$$EP = \frac{E_{\text{heat}} \times 1.8/F_{\text{geo}} + E_{\text{DHW}} \times 1.8 + E_{\text{property}} \times 1.8}{A_{\text{temp}}} = 108.3 \text{ kWh/(m}^2\text{·year)}. \quad (11)$$

Of that, space heating is 36, domestic hot water is 45, and property electricity is 27, all in kWh/m^2·year. The heat pump divides the space-heat bill by three, so the hot-water cylinder, still a plain electric heater, weighs more in the rating than the heating does.

The new-building limit for a multi-family house is 75 kWh/m^2·year. There is no ventilation supplement: the specific flow is under 0.35 L/(s·m^2). The ratio is 1.44, which is energy class E (Figure 4).

Class C means as good as the requirement for a new building. Classes toward A are better. This building does not meet that requirement. Most of the primary energy is electricity, and electricity is weighted at 1.8. The same heat on district heating (weight 0.7, no COP) would look different. The birth date fixes the system, so that comparison is not the result, only a reminder that the class is about the carrier as well as the heat.

One limit of the COP: it is taken as 3 even on the design day. A real air-to-air pump loses capacity in a cold spell. The design power above is the heat the building needs. The pump may need a backup heater to deliver it at DVUT. That backup is not in the annual COP of 3, so the annual electricity, and the class, are a little optimistic.

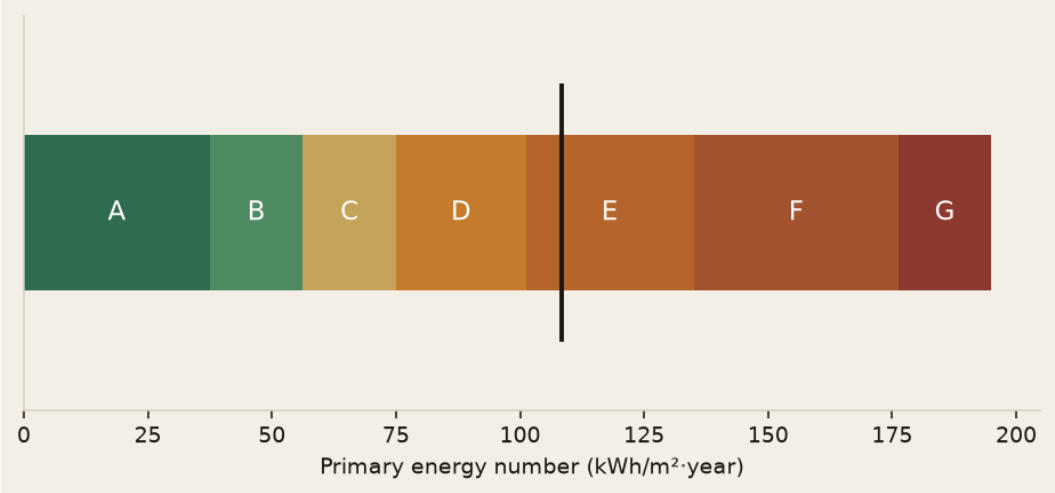


Figure 4: Primary energy against the BBR class boundaries for a new multi-family building. The result sits in class E.

10 What would actually cut the heating

Each line in Table 4 repeats the normal-year correction with one change applied.

Table 4: Measures, same climate correction as the base case.

Measure	Heat (MWh)	Saved (MWh)	Class
Triple glazing, $U = 1.0$ W/(m^2·K)	56.2	25.5	D
Heat recovery, 80 % of exhaust air	52.6	29.2	D
Balcony break, $\Psi = 0.20$ W/(m·K)	74.2	7.6	E
All three together	22.1	59.6	D

The order of usefulness, on this building:

1. **Heat recovery on the exhaust air.** The ventilation loss is large and today it is thrown away. An exhaust-air heat pump, or a supply-and-exhaust unit with 80 % recovery, attacks that term directly. Leakage is left unchanged.
2. **The windows.** $U = 2.9$ on more than 100 m^2 is the biggest single transmission term. Triple glazing around $U = 1.0$ is the obvious replacement.
3. **The balcony slabs.** $\Psi = 0.6$ W/(m·K) over the long façades is a serious bridge, which the brief already flags. A thermal break at the slab edge cuts it without touching the apartments much.

Doing all three still leaves domestic hot water and property electricity, both counted as electricity at factor 1.8. Together those two terms are about 72 kWh/m^2·year before any space heat is added. The class moves from E to D, but the envelope alone does not reach class C, because the primary-energy number is not only the space heating.

11 What this does not claim

- The city list was not inside the PDF. City 29 is Danderyd because the SMHI place list, without Abisko, puts Bjurholm at number 14 (a known reading of the same brief) and Danderyd at number 29.
- The solar chart in the brief was read by eye. A few watts per square metre either way does not change the class; it does move the solar megawatt-hours.
- Internal gains other than the constant 100 W per person and the appliance wattage are not modelled. No night setback, no shutters.
- The 10 cm active-depth rule is used only to pick DVUT. The annual energy calculation is steady-state, hour by hour, not a dynamic simulation.
- Hot water and property electricity are Sveby standard values, not measurements from this association.
- The mean outdoor temperature in 2017 was 7.65 °C, against 7.32 °C in the normal year, so 2017 was slightly milder. The coldest hour of 2017 (−17.1 °C) is colder than the 1-day DVUT (−13.5 °C). DVUT is a sustained spell, not the coldest hour.

References

[1] Assignment brief, course ETA013, University of Gävle. Deadline 17 November 2026.

[2] Boverket / SMHI, DVUT 1991–2020, open data. Place Danderyd, ortnr 102646.

[3] Sveby climate file for 2017, version 2.0, place Danderyd (102646).

[4] Sveby typical year 1991–2020, same place. Composite year: the month and hour columns are used, not the source year.

[5] Figure 1 of the brief: Nevander and Elmarsson, *Fukthandboken*, 1994. Monthly solar irradiance at Örebro, 60° N.

[6] Boverket, BBR: electricity weighting factor 1.8, new multi-family limit 75 kWh/m^2·year, classes A–G. $F_{\text{geo}} = 1.0$ for Danderyd.

[7] Sveby user input data for multi-family buildings: domestic hot water 25 kWh/m^2·year, property electricity 15 kWh/m^2·year.

[8] ISO 13790, 10 cm active depth for the thermal mass used to select DVUT.